



ENVT. MGT. Department
SIMHADRI



Ref:SMTPP/ENVT. MGT./2022-23/EMG-AU/742891

Date:

07/01/2023(10:04:24)

Subject:Approval of LMI-EMG-Measurement of Particulate and Gaseous Emissions From Simhadri Super Thermal Power Station

The following LMI pertaining to Simhadri-EMG is submitted hereby for approval,

LMI/OGN/ENV/002 REV.7/ -**Measurement of Particulate and Gaseous Emissions From Simhadri Super Thermal Power Station**

Revision of OGN is envisaged in this revision , proposed for approval of revision. LMI is enclosed for reference.

Submitted for approval please.

Dr Pradeep M

Manager-EMG

Submitted by:

प्रदीप एम - प्रबंधक

Pradeep Muthiah - 027524 (MANAGER ,ENVT. MGT.) ,Simhadri
(PRADEEPM@NTPC.CO.IN, 9445422004)

On:07/01/2023(10:04:24) **Ref:**SMTPP/ENVT. MGT./2022-23/EMG-AU/742891

Revisions marked in yellow in the LMI doc. Revised LMI is put up for review and approval please.

Submitted by:

डॉ वी जयन - अपर महाप्रबंधक

Dr Velayudhanpillai Jayan - 006202 (ADDL. GENERAL MANAGER ,ENVT. MGT.)
(VJAYAN@NTPC.CO.IN, 9650991721) ,Simhadri

On:07/01/2023(10:41:59) **Ref:**SMTTP/ENVT. MGT./2022-23/EMG-AU/742891

may pl. comment

Submitted by:

पीयूष श्रीवास्तव - अपर महाप्रबंधक

Piyush Srivastava - 006316 (ADDL. GENERAL MANAGER ,CHEMISTRY)
(PIYUSHSRIVASTAVA@NTPC.CO.IN, 9650999535) ,Simhadri

On:07/01/2023(19:36:23) **Ref:**SMTTP/ENVT. MGT./2022-23/EMG-AU/742891

LMI Reviewed. Remarks given against Chemistry dept's points.

Points with Chemistry/EMG responsibility

Remark

The measurement of emissions should be carried out in accordance with the requirements detailed in standards / procedure prescribed by CPCB/ APPCB. Station may also get the same work done through EPA recognized third party using cascade impactor having cut off size as mentioned above at least three particle size below 10.0 μ m and 2/3 above 10.0 μ m

MoEF Certified laboratory is appointed for 3rd party monitoring. Contract period is from Nov-2022 to Oct-2024.

Each unit shall be sampled once a fortnight. Manual sampling and analysis shall be done by Chemistry department internally or through a third party (EPA recognized by MoEF&CC/ State Pollution Control Board recognized Labs).

In Practice

A report of the manual measurement, calibration

details and inter-comparison data shall be issued by

the Head of Chemistry/ Env. Monitoring Group responsible for carrying out the measurements. This should include all the additional parameters given in results reporting data sheets enclosed, including remarks for abnormal results.

Shall be complied from Jan-2023.

May pl forward.

Submitted by:

लवण्या कलत्थुरु - उप प्रबंधक (रसायन)

Lavanya kalatthuru - 103064 (DEPUTY MANAGER(CHEMIST) ,CHEMISTRY)
(LAVANYAK@NTPC.CO.IN, 7382621063) ,Simhadri

On:09/01/2023(16:30:13) **Ref:**SMTTPP/ENV.T. MGT./2022-23/EMG-AU/742891

Submitted by:

पीयूष श्रीवास्तव - अपर महाप्रबंधक

Piyush Srivastava - 006316 (ADDL. GENERAL MANAGER ,CHEMISTRY)
(PIYUSHSRIVASTAVA@NTPC.CO.IN, 9650999535) ,Simhadri

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Submitted for review and approval

Submitted by:

प्रदीप एम - प्रबंधक

Pradeep Muthiah - 027524 (MANAGER ,ENV.T. MGT.)
(PRADEEPM@NTPC.CO.IN, 9445422004) ,Simhadri

On:10/01/2023(10:13:51) **Ref:**SMTTPP/ENV.T. MGT./2022-23/EMG-AU/742891

Pl review

Submitted by:

टी.आर. राजगोपालन् - अपर महाप्रबंधक

TR Rajagopalan - 003994 (ADDL. GENERAL MANAGER ,C & I MAINT)
(TRRAJAGOPAL@NTPC.CO.IN, 9440918024) ,Simhadri

On:10/01/2023(17:26:38) **Ref:**SMTTP/ENV.T. MGT./2022-23/EMG-AU/742891

Responsibilities pertaining to C&I noted for continuous compliance and put up for further approval.

Submitted by:

जी एच राव - अपर महाप्रबंधक

G H Rao - 006589 (ADDL. GENERAL MANAGER ,C & I MAINT- BOILER)
(GHNRAO@NTPC.CO.IN, 8331016245) ,Simhadri

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Reviewed and forwarded for further approval

Submitted by:

टी.आर. राजगोपालन् - अपर महाप्रबंधक

TR Rajagopalan - 003994 (ADDL. GENERAL MANAGER ,C & I MAINT)
(TRRAJAGOPAL@NTPC.CO.IN, 9440918024) ,Simhadri

On:11/01/2023(09:00:13) **Ref:**SMTTP/ENV.T. MGT./2022-23/EMG-AU/742891

May please be approved.

Submitted by:

Dhurjati Prasad Patra - 004573 (GENERAL MANAGER ,O & M)
(DPPATRA@NTPC.CO.IN, 8275045446) ,Simhadri

On:12/01/2023(11:08:32) **Ref:**SMTTP/ENV.T. MGT./2022-23/EMG-AU/742891

Approved & Submitted by:

संजय कुमार सिन्हा - महाप्रबंधक

Sanjay Kumar Sinha - 005185 (GENERAL MANAGER ,HEAD OF PROJECT)
(SKSINHA06@NTPC.CO.IN, 9431215239) ,Simhadri

On:12/01/2023(15:18:26) **Ref:**SMTTP/ENV.T. MGT./2022-23/EMG-AU/742891

LOCATION MANAGEMENT SYSTEM

LMI/OGN/ENV/002

Rev. No.: 7

**Measurement of Particulate and Gaseous
Emissions From Simhadri Super
Thermal Power Station**

Approved for Implementation by

HOP (Simhadri)

Date:

LMI APPROVAL DOCUMENT

1. TITLE OF LMI	MEASUREMENT OF PARTICUALTE AND GASEOUS EMISSIONS FROM SIMHADRI SUPER THERMAL POWER STATION
2. a) DOCUMENT CODE b) REVISION/ISSUE NO. & DATE	LMI/OGN/ENV/002 REV.7/ 07.01.2023
3. ORIGINAL/REF. DOCUMENT	COS-ISO-00-OGN/ENV/002 Rev.No.:3 Dec 2022
4. REVISION MADE BY	Dr Pradeep M, Manager (EMG &AU)
5. CHECKED BY	Dr V Jayan, AGM (EMG& AU)
6. REVIEWED BY	
	HOD (Chem)
	HOD (C&I)
	GM(OPN)
7. DATE OF ISSUE	07.01.2023

**MEASUREMENT OF PARTICULATE AND GASEOUS EMISSIONS
FROM SIMHADRI SUPER THERMAL POWER
STATION**

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MEASUREMENT OF PARTICULATE AND GASEOUS EMISSIONS FROM SIMHADRI STPS

1.0 INTRODUCTION

This LMI specifies the procedures for the measurement of particulate and gaseous emissions from Simhadri Power Station.

Such measurements are necessary to provide an absolute value besides direct reading on-line/continuous/portable dust and gas monitors, of the particulate emission, to demonstrate compliance against a specific standard or the requirements of consent orders by the Pollution Control Authority

2.0 SUPERSEDED DOCUMENTS

LMI/OGN/ENV/002 REV. 6

3.0 SCOPE

This LMI applies to the measurement of particulate and gaseous emissions (SPM, SO₂ and NO_x) from Simhadri STPS.

4.0 PROCEDURES

The measurement of emissions should be carried out in accordance with the requirements detailed in standards / procedure prescribed by CPCB/ APPCB. Station may also get the same work done through EPA recognized third party using cascade impactor having cut off size as mentioned above at least three particle size below 10.0μ and 2/3 above 10.0 μ.

All the particulate emission, SO₂ and NO_x emissions results shall be corrected to the reference conditions of STP (25°C, 1 bar), dry and 6% O₂. In addition, all relevant plant operating conditions during the emission measurements are to be logged as outlined in the results reporting data sheets shown in Appendix II and III.

Res: HOD(Chem)/HOD(EMG)

5. FREQUENCY OF SAMPLING

Manual Sampling and Analysis: Each unit shall be sampled once a fortnight. Manual sampling and analysis shall be done by Chemistry department internally or through a third party (EPA recognized by MoEF&CC/ State Pollution Control Board recognized Labs).

Res: HOD(Chem)

CEMS: Real time emission data being monitored by online CEMS for particulate, SO₂ and NO_x shall be made available online and is to be transmitted to the regulators on real-time basis as per the requirements of the regulator. Record of calibration of all the analyzers, inter comparison of the CEMS data with manual monitored data shall be properly maintained. Remote Calibration shall be opted as stipulated by regulator (APPCB/CPCB). The attached format in Annexure-I&II are to be adopted.

Res: HOD(C&I)/HOD(EMG)

6. CONDITIONS FOR MEASUREMENTS

The plant operating conditions for the emission measurement is arranged and maintained by the Station.

Emissions should not normally be measured during periods of cleaning, dusting, or soot blowing; hence these operations should be carried out before the measurements are taken to allow sufficient time for the later to be completed.

Res: HOD (O)/HOD(Chem)

7. EQUIPMENT

7.1 The equipment normally used shall be a stack monitoring kit for SPM, SO₂, NO_x and Hg. Other equipment may be used provided it has been demonstrated to be adequate for the purpose. The equipment uses fine grade double thickness alundum/ cellulose/ glass fibre/ asbestos thimble or filter medium capable of collection of particulates down to 0.3 micron as specified in Emission Regulation, Part-III and Laboratory Analytical Techniques Series: LATS / 80 / 2013-2014 Guidelines on Methodologies for Source Emission Monitoring issued by CPCB.

Res: HOD(Chem.)

7.2 In addition to manual equipment Simhadri Power station is provided with online measuring equipment for the measurement of SPM, SO₂ and NO_x in the flue gasses.

The online equipment is being calibrated every month and daily average readings are available in MMI system

Res: HOD(C&I)

8. SAMPLING PORTS

The sampling ports are provided at a height of 88 m in all four unit Flues and working platforms are constructed as stipulated by AP Pollution control board in their NOC. Power supply arrangement is also provided as per stipulations.

Res: HOD (O&M)/HOD (EMG)

9. INTER-COMPARISON OF CEMS AND MANUAL MONITORING DATA

During inter comparison of manual monitoring data and CEMS data, If there is difference of more than 20 % , calibration of the both manual monitoring equipments and CEMS is taken up.

Res: HOD (Chem)/HOD (EMG)

10. PRESENTATION OF RESULTS

A report of the manual measurement, calibration details and inter-comparison data shall be issued by the Head of Chemistry/ Env. Monitoring Group responsible for carrying out the measurements. This should include all the additional parameters given in results reporting data sheets enclosed, including remarks for abnormal results.

Res: HOD(Chem)/HOD(EMG)

11. REPORTING

The report in prescribed format shall be furnished every month to

- i) Head of Station/Project
- ii) Head, EMG, Regional HQ
- iii) Head, Environment Group, Corporate Centre.
- iv) Uploading of data in prescribed format in Parivesh.

12. REFERENCE DOCUMENTS

- 1) Emission Regulation Part-III (December 1985), Central Pollution Control Board, New Delhi.
- 2) Operation and maintenance manual of CEMS
- 3) Environment (Protection) Amendment Rules 2015
- 4) Air (Prevention & control of pollution) Act, 1981
- 5) Air (Prevention & control of pollution) Rules, 1982

- 6) The Environment (Protection) Rules, 1986
- 7) NOC/EC/CTO of Simhadri STPS
- 8) Directive by CPCB for installation of CEMS
- 9) Guidelines for continuous emission monitoring system by CPCB Aug'2018.

13.0 REVIEW

The HOD (EMG), CC will be responsible for reviewing this document in every two years or as necessary.

Appendix – I

ONLINE STACK MONITORING SYTEM

Name of the Station: Simhadri STPS

Manual Monitoring Frequency	: Once in Fortnight for every unit
Last Calibration of CEMS	:
Comparison with Manual Monitored data	:
Online percentage availability of CEMS data in current FY	:
Online data connectivity status to SPCB/CPCB	:
Auto diagnostic system access to SPCB/CPCB	:
Sulphur content in Coal(analyzed on)	:
Mercury content in Coal(analyzed on.....)	:
Exception in the CEMS availability, if any	:
Name of the Person responsible for CEMS	:
Contact Number &E-mail D	:
Name of the person & his contact detail communicated to CPCB/APPCB/CC on (Date)	:

RECORD OF STACK EMISSION FOR THE MONTH : YEAR

Sl No	Date & Time	Unit No	Load (MW)	Flue gas parameters						No of Oil guns in Service	Sp Coal Cons. (kg/ KWh)	% Av Ash in Coal as fired	Parameter	Concentration (mg/Nm ³)
				FG Pass monitored A/B/C or common duct	AV FG Vel. (Km/hr)	FG Tem p. (°C)	Measured AV FG Flow rate (Nm ³ /hr)	Design Flue gas Flow (Nm ³ /hr)	Flue Gas % O ₂ Concentration					
													Particulate matter	
													SOx (6%O ₂)	
													NOx ((6%O ₂))	

Appendix-II E.S.P

STATION :

MONTH & YEAR:

ESP DATA TO BE RECORDED AT THE TIME OF PARTICULATE MEASUREMENT

SI No	Date & Time	Unit	ESP No	Fields		Voltage (KV)		Spark rate (per min)		Remarks
				T.I	I.S	In 1st row fields	In last row fields	In 1st row fields	In last row fields	

Legend T.I. - Total Installed I.S. - In Service C.E. - Collecting electrode

NOTES

1. Please specify all data for the entire month for Unit-1 first, then for Unit-II and so on.
2. If 1st and/or last rows are not in service, then please give data for voltage and spark rate for next service row (please indicate row no. in remarks column in this case.)